MTCFD Manual

User Manual for Melt Pool and Evaporation CFD Simulation in Metal Additive Manufacturing

Prepared for metal additive manufacturing CFD simulation projects. This manual documents the computational fluid dynamics simulation framework developed to predict melt pool geometry and dynamics during laser scanning, as well as macroscopic defect formation and composition changes caused by evaporation. The developed simulation method couples a thermal-fluid model with an evaporation model, enabling the consideration of recoil pressure, latent heat, selective evaporation, and near-surface composition variation, and allowing prediction of final composition changes in multicomponent alloy systems.

Project / Repository	[MTCFD]
Solver / Code Base	[OpenFOAM-based CFD solver]
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Prepared by	[Matae Lee / Postech CMSE lab / MSE]

Recommended use:

- Use this document as the main user manual for simulation setup, execution, and post-processing.
- Refer to this manual for model background, module descriptions, and example cases for metal additive manufacturing CFD simulations.

Contents

1. Introduction 3
A. Purpose of the solver 3
B. Scope and main features 3
2. Installation 5
A. Requirements 5
B. Solver compilation 5
C. Directory structure 5
3. Examples 7
A. Basic example 7
B. Parameter variation example 8
C. Post-processing and visualization 12
4. References 14

1. Introduction

This section should explain the purpose of the CFD code, the intended users, the physics scope, and the overall workflow from case preparation to numerical results. Keep the opening concise and practical so a new user can understand what the solver is for before reading implementation details.

A. Purpose of the solver

Additive manufacturing (AM) processes involving laser irradiation generate highly localized heating, rapid melting and solidification, and intense evaporation at the melt pool surface. These phenomena strongly affect melt pool geometry, keyhole behavior, defect formation, and final composition, particularly in multicomponent alloys containing volatile elements. Accurate prediction of these behaviors is therefore essential for understanding and optimizing AM processes.

The purpose of this solver is to predict the formation and evolution of the melt pool during laser-based AM by coupling a thermal-fluid model with an evaporation model. The solver is designed to simulate not only melt pool shape and fluid flow, but also evaporation-induced recoil pressure, latent heat effects, and composition change in multicomponent alloy systems. Through this coupled approach, the tool enables quantitative analysis of keyhole formation, selective evaporation, and final compositional variation in AM products.

This solver was developed to provide a practical computational tool for analyzing AM melt pool behavior under various process conditions such as laser power, scan speed, and alloy composition. It is particularly intended for cases where evaporation plays a significant role in determining process stability, keyhole dynamics, and composition redistribution.

B. Scope and main features

This solver focuses on laser-based metal AM processes in which melt pool dynamics and evaporation must be considered simultaneously. It applies a three-dimensional computational fluid dynamics framework to calculate temperature evolution, melt flow, free-surface deformation, and surface-force-driven phenomena during laser irradiation. The thermal-fluid part is implemented on the basis of OpenFOAM and solves the governing transport equations using a finite volume framework.

A key feature of the solver is the coupling between the thermal-fluid model and the evaporation model. The thermal-fluid model provides surface temperature, melt pool geometry, and local composition information to the evaporation model. In return, the evaporation model supplies evaporation flux, recoil pressure, and latent heat loss back to the thermal-fluid solver. This two-way coupling allows the simulation to reflect the dynamic interaction between melt pool evolution and evaporation phenomena.

The solver also includes several advanced physical capabilities. Laser energy absorption at the evolving melt pool surface is evaluated using a ray-tracing approach, enabling more realistic treatment of absorptivity under changing keyhole geometry. Multicomponent composition transport near the free surface is handled using a volume-of-fluid-based formulation, which makes it possible to track selective evaporation and local composition variation. In addition, the evaporation model incorporates thermodynamic effects such as saturated vapor pressure, activity-based multicomponent behavior, and temperature-dependent evaporation kinetics.

The current solver is intended for the simulation and analysis of melt pool morphology, keyhole behavior, and evaporation-induced composition change in AA7075 during laser-based additive manufacturing. At

its current stage, the solver is configured specifically for AA7075 and is mainly used for baseline simulations, process-condition comparison, and validation against experimental observations for this alloy system. Therefore, the present version of the solver provides a practical platform for studying defect formation, melt pool instability, and process-dependent composition change in AA7075.

2. Installation

Use this section to explain the software modules in the same order that users encounter them. A stable module section is especially helpful when the codebase grows and different contributors touch pre-processing, solver logic, and analysis tools.

A. Requirements

- OpenFOAM Foundation version 10 is required.
  The OpenFOAM-10 source code is available at:
  https://openfoam.org/download/10/

- The laserbeamFoam solver developed by Flint et al. must be installed.
  The original laserbeamFoam implementation is available at:
  https://github.com/laserbeamfoam/LaserbeamFoam.git
  In this work, the src/laserHeatSource module from the original laserbeamFoam package is used as the laser heat source model.

- This solver runs in a Linux environment.

B. Solver compilation

The solver directory contains source code for compiling the laserbeamFoam-based solver [1,2].

Before compilation, the include path to the `laserHeatSource` library installed with laserbeamFoam must be specified in the solver build options.

Specifically, the path to `laserHeatSource/lnInclude` should be added to the `EXE_INC` entry in the `Make/options` file.

The correct path can be identified from the local laserbeamFoam installation.

1. Navigate to the solver directory:

```
cd solver
```

2. Compile the solver:

```
./Allwmake
```

After successful compilation, the solver can be executed using:

```
laserbeamFoam
```

C. Directory structure

The present solver repository is organized into directories for case execution and solver source code. A simplified directory structure is shown below.

```
Repository/
├─── run_scanning/      : Example case directory for running scanning simulations.
│   ├─── constant/      : Contains material properties, laser settings, and process parameters.
│   ├─── initial/       : Initial field files used at the start of the simulation.
```

```
│   ├── system/      : Contains simulation time settings, numerical schemes, and solver control parameters.
│   ├── open.foam      : ParaView/OpenFOAM file for post-processing and visualization.
│   └── slurm_multi      : Batch script for running cases on a cluster with Slurm.
├── solver/      : Source code directory of the customized solver.
│   ├── Make/      : Build configuration files for compilation.
│   ├── Allwmake      : Shell script for compiling the solver and related files.
│   ├── laserbeamFoam.C      : Main source file of the solver.
│   ├── evaporation.H      : Header file related to evaporation modeling.
│   ├── alphasEqn.H      : Header file for phase-fraction transport equations.
│   └── ...      : Additional source and header files used by the solver.
├── LICENSE      : License information for the repository.
└── README.md      : Overview, usage instructions, and basic repository information.
```

3. Examples

This section presents the example case included in the repository, run_scanning.
This case is provided to help users understand the basic workflow of the solver, from case setup to execution and post-processing.
By following this example, users can check the required input files, run procedure, and main output results.
Since it is the main example included in the repository, it can also be used as a starting point for further modifications and applications.

A. Basic example

1. Generate the computational mesh:

```
blockMesh
```

2. Initialize fields:

```
cp -r initial/ 0/
setFields
```

3. Decompose the domain for parallel execution:

```
decomposePar
```

4. Solver execution

- Serial execution:

```
laserbeamFoam
```

- Parallel execution (example using 32 processors):

```
mpirun -np 32 laserbeamFoam -parallel >& log
```

B. Parameter variation example

This example shows how the simulation conditions can be changed by modifying input files in the constant folder. By adjusting the laser settings, process conditions, and material properties, the same case structure can be used to represent different processing conditions and even different alloy systems. This makes the case suitable for simple parameter studies without changing the main solver or case setup.

1. LaserProperties

The laser conditions can be modified in the LaserProperties file in the constant folder. This file contains the main laser-related parameters used in the simulation, such as the laser radius, wavelength, electron number density, and other beam settings. By changing these values, the user can control the characteristics of the applied heat source.

In particular, the laser power and scanning motion can be adjusted through the timeVsLaserPower and timeVsLaserPosition input files, respectively. The timeVsLaserPower file defines how the laser power changes with time, while the timeVsLaserPosition file specifies the laser position as a function of time. Therefore, the scanning speed can be controlled by changing the position data with respect to time. This structure allows flexible definition of both constant and time-dependent laser conditions within the same simulation case.

```
timeVsLaserPosition
{
    file     "$FOAM_CASE/constant/timeVsLaserPosition";
    outOfBounds clamp;
}                                   Laser scan speed

timeVsLaserPower
{
    file     "$FOAM_CASE/constant/timeVsLaserPower";
    outOfBounds clamp;              Laser power
}


laserRadius 40.00e-6;               Laser radius


N_sub_divisions 1;


wavelength       1.064e-6;
e_num_density    2.26e29;           Related to absorptivity


Radius_Flavour  2.0;     Beam radius defined at the 1/e^2
```

2. physicalProperties.metal

The material properties can be modified in the physicalProperties.metal file in the constant folder. This file defines the main thermophysical properties required for the simulation, including viscosity, density, thermal conductivity, heat capacity, liquidus temperature, solidus temperature, latent heat of melting, and electrical resistivity. In particular, viscosity, thermal conductivity, and heat capacity are given as temperature-dependent properties, allowing the simulation to reflect their variation over the relevant temperature range. By changing these values, the same simulation framework can be used to represent different materials and alloy systems.

```
nu table                                  Viscosity
(
    (869 5.87866E-07)
    (1501.67 3.52427E-07)
    (2134.33 3.01095E-07)
    (2767 2.89045E-07)
);

rho 2409.86;                              Density

kappa table                               Thermal conductivity (liquid)
(
    (869.00 65.86)
    (2767.00 105.24)
);

kappas table                              Thermal conductivity (solid)
(
    (298.10 162.05)
    (375.99 165.71)
    (830.00 135.26)
);


cp table                                  Heat capacity (liquid)
(
    (869.00 1188.00)
    (2767.00 928.94)
);

cps table                                 Heat capacity (solid)
(
    (297.75 977.95)
    (830.00 1195.24)
);

Tsolidus 806.0;                           Liquidus temperature
Tliquidus 925.0;                          Solidus temperature
LatentHeat 494000;                        Latent heat of melting
beta 0.000025;

elec_resistivity 2.40e-7;                 Related to absorptivity
```

3. phaseProperties

The phase-related properties can be modified in the phaseProperties file in the constant folder. This file includes the surface tension, its temperature dependence, the vaporization temperature, molar mass, and the latent heat of vaporization. In particular, the surface tension can be defined as a temperature-dependent property, allowing the model to account for its variation with temperature. By changing these values, the user can adjust the phase-change and interfacial behavior used in the simulation.

```
sigma                    Surface tension
{
    type temperatureDependent;
    T T;
    sigma table
    (
        (869 0.8525674)
        (2767 0.3089802)
    );
}

dsigmadT    -0.0002864;   Surface tension
                          T dependence

p0      100000.0;
Tvap 2767.0;              Vaporization temperature
Mm 2.698e-2;              Molar mass
LatentHeatVap 10.5e6;     Latent heat of vaporization
```

4. Evaporation model properties

In the evaporation model, the relevant material properties are defined in createFields.H. The current implementation already includes the data for the Al–Cu–Mg–Zn system, but if a different alloy system is to be used, these values should be modified accordingly before compilation. This file contains the properties required for each element, including the latent heat of vaporization, vaporization temperature (T_vap), molar mass, density (rho), and weight fraction. Since these values are assigned on an element-by-element basis, the model can represent the evaporation behavior of each alloying element individually.

By defining the weight fractions, the initial composition of the sample can be prescribed. This makes it possible to simulate how different initial alloy compositions influence melt pool morphology as well as composition change during processing. In other words, the model can be used not only to track evaporation-induced compositional variations, but also to examine how the starting alloy chemistry affects the overall melt pool behavior.

At present, the example setup is configured for the Al–Cu–Mg–Zn system corresponding to 7075 alloy. However, there are plans to extend the implementation in the future so that additional alloying elements can be incorporated.

```
// 4. properties
// must fix this part
double Latentheatgas_list[4] = {9462000.0, 4720995.0, 5241719.8, 1754000};
double Tvap_list[4] = {2790, 2835, 1363, 1180};
double molarmass_list[4] = {0.02698, 0.063546, 0.024305, 0.06538};
double rho_list[4] = {2698.9, 8000, 1590, 6580};
double wf_list[4] = {0.9015, 0.0152, 0.0244, 0.0589};
scalar air_molarmass = 0.039948;
```

The evaporation calculation also takes thermodynamics into account. For this reason, thermodynamic parameters are required in addition to the material properties defined for each element. In the present example, the system is assumed to be a quaternary Al–Cu–Mg–Zn alloy, and the evaporation behavior is evaluated using the liquid-phase thermodynamic parameters for the relevant binary and ternary subsystems. These parameters are listed below.

To perform calculations for a different alloy system, both sets of inputs must be updated accordingly: the element-specific properties described above, such as latent heat of vaporization, vaporization temperature, molar mass, density, and weight fraction, as well as the thermodynamic parameters used for the liquid phase. In other words, changing the target alloy composition requires not only redefining the physical properties of the constituent elements, but also replacing the corresponding binary and ternary thermodynamic parameters so that the evaporation calculation remains consistent with the selected alloy system.

Binary

```
// Al Cu Mg Zn
// Al Cu
FL1[0][0][1] = -67094;
FL1[1][0][1] = 8.555;
FL1[2][0][1] = 32148;
FL1[3][0][1] = -7.118;
FL1[4][0][1] = 5915;
FL1[5][0][1] = -5.889;

// Al Mg
FL1[0][0][2] = -12000;
FL1[1][0][2] = 8.566;
FL1[2][0][2] = 1894;
FL1[3][0][2] = -3;
FL1[4][0][2] = 2000;

// Al Zn
FL1[0][0][3] = 10465.55;
FL1[1][0][3] = -3.39259;

// Cu Mg
FL1[0][1][2] = -36984;
FL1[1][1][2] = 4.7561;
FL1[2][1][2] = -8191.29;

// Cu Zn
FL1[0][1][3] = -40688.6;
FL1[1][1][3] = 12.8655;
FL1[2][1][3] = 2252.3;
FL1[3][1][3] = -4.8069;
FL1[4][1][3] = 5405.9;
FL1[5][1][3] = -2.9197;

// Mg Zn
FL1[0][2][3] = -77729.24;
FL1[1][2][3] = 680.52266;
FL1[2][2][3] = 3674.72;
FL1[3][2][3] = 0.57139;
FL1[4][2][3] = -1588.15;
```

Ternary

```
// Al Cu Mg - X

// Al Cu Zn
TL1[0][0][1][3] = 0;
TL1[1][0][1][3] = 0;
TL1[2][0][1][3] = 5475;
TL1[3][0][1][3] = 13000;
TL1[4][0][1][3] = 43800;
TL1[5][0][1][3] = 0;
TL1[6][0][1][3] = 0;
TL1[7][0][1][3] = 0;

// Al Mg Zn
TL1[0][0][2][3] = 0;
TL1[1][0][2][3] = 0;
TL1[2][0][2][3] = -11475;
TL1[3][0][2][3] = -11475;
TL1[4][0][2][3] = -11475;
TL1[5][0][2][3] = 11;
TL1[6][0][2][3] = 11;
TL1[7][0][2][3] = 11;

// Cu Mg Zn - X
```

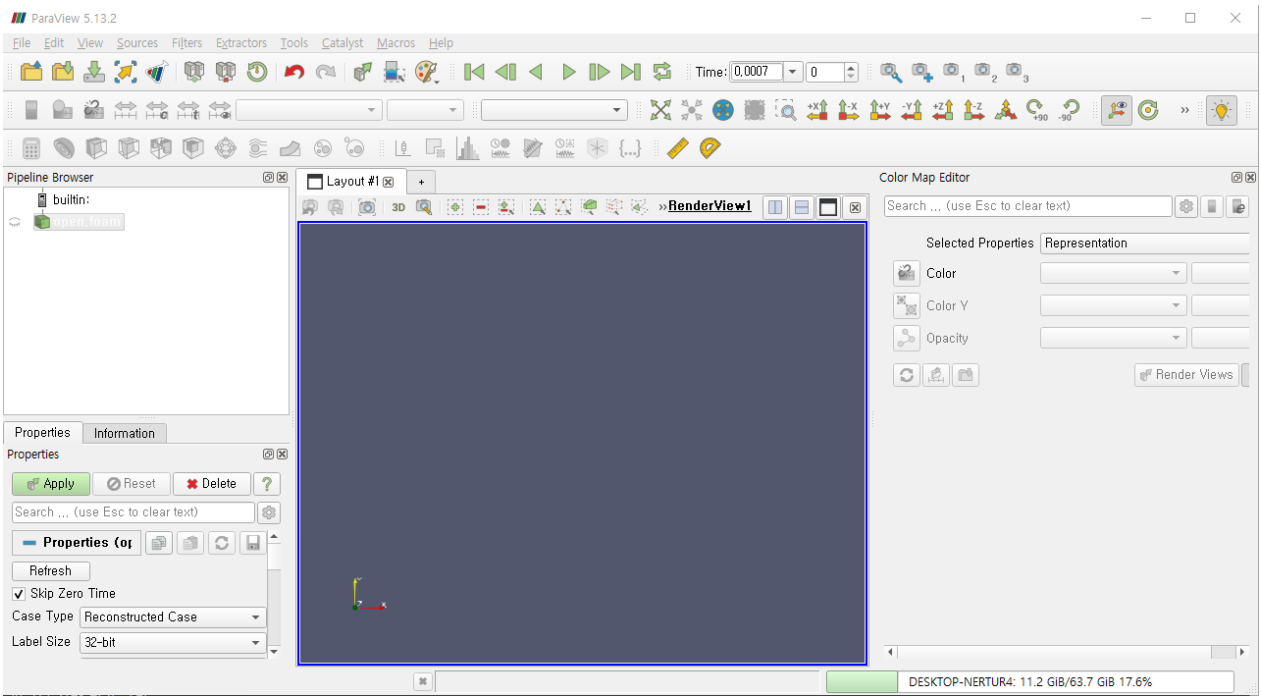
C. Post-processing and visualization

After a parallel run, the decomposed results should first be combined into a single dataset using reconstructPar. This step reconstructs the fields from the processor directories and gathers them into one case, making the simulation results easier to inspect and process further.

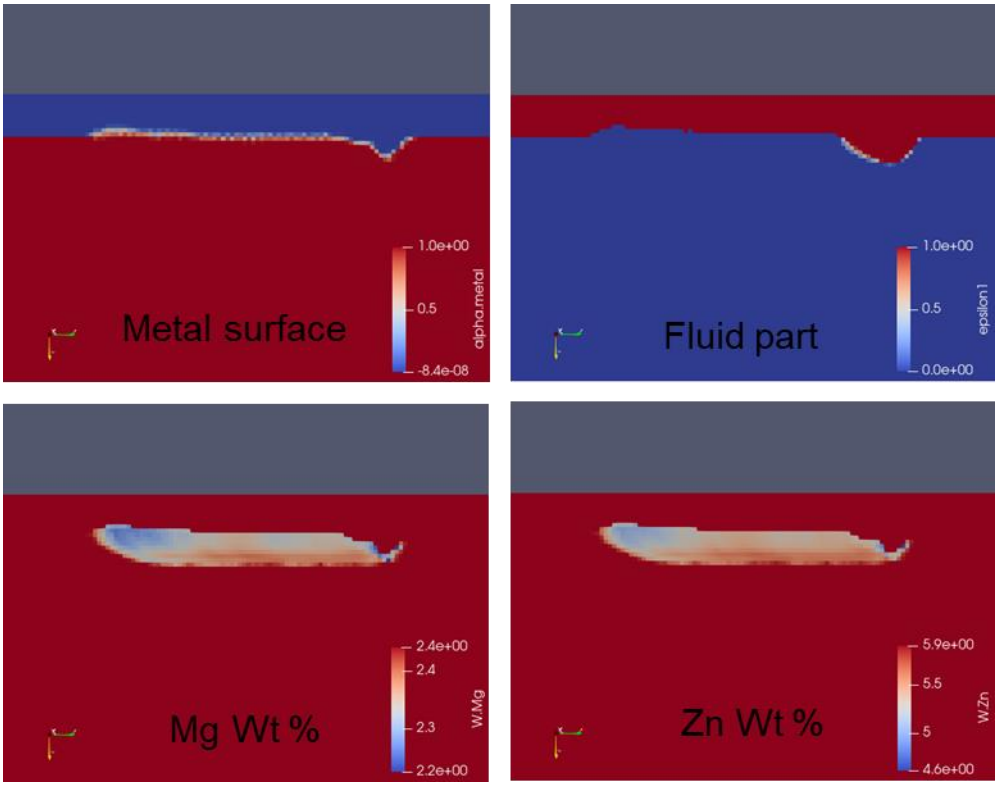
```
reconstructPar
```

Once the reconstruction is completed, the results can be opened and visualized in ParaView. ParaView is commonly available together with an OpenFOAM installation, so it can usually be used directly after OpenFOAM is installed. This allows the user to examine the melt pool shape, temperature field, velocity field, and other simulation outputs in a convenient graphical environment.

1. Open ParaView and load the open.foam file.



2. Click Apply and inspect the results.



4. References

[1] T.F. Flint, J.D. Robson, G. Parivendhan, P. Cardiff, laserbeamFoam: Laser ray-tracing and thermally induced state transition simulation toolkit, SoftwareX 21 (2023) 101299. https://doi.org/10.1016/j.softx.2022.101299.

[2] T.F. Flint, J.D. Robson, P. Esmati, N. Grilli, G. Parivendhan, P. Cardiff, Version 2.0 — LaserbeamFoam: Laser ray-tracing and thermally induced state transition simulation toolkit, SoftwareX 25 (2024) 101612. https://doi.org/10.1016/j.softx.2023.101612.